

Math 120 Spring 2024 - Final Preparation

(Spring 2019) 1. Find $\frac{d^2}{dx^2}[\ln(x^2 + 2)]$. DIY. (Sanity check: The first derivative should have been $\frac{2x}{x^2+2}$.)

(Spring 2019) 4b. An epidemic of measles has started in a city, and the number of people infected is modeled by the function $P(t) = \frac{1000}{1+19e^{-0.3t}}$, where t is measured in days since the beginning of the epidemic. At what rate is the epidemic spreading after 3 days?

Extra Q: What is the carrying capacity (maximum number of infected people) in this model? What is the number of infected people at the beginning in this model?

(Spring 2018 Washington) 4. Aristotle is a gardener who wants to build a rectangular garden. He has \$700 to spend on a fence for this garden. The side closest to the road requires a special fence that costs \$3 per foot, whereas the rest of the sides will be built using a fence that costs \$1 per foot. What are the dimensions of the biggest area that Aristotle can build?

(Spring 2018 Washington) 5 (a). Benazir is driving a car on a straight road. After t seconds she is $f(t) = t^3 + 2t^2$ meters away from her starting position. What is her velocity after 2 seconds? (No need to specify the units.)

Extra Q: Now forget $f(t)$. If you know that her velocity is $v(t) = 3t^2 + 4t$, can you recover her distance function $f(t)$? Notice that the distance traveled at time $t = 0$ is 0, so $f(0) = 0$.

(Spring 2018 Washington) 8(b). for $f(x, y) = x^2 + 4xy + 3y^2$, the point $(0, 0)$ is a critical point. Does $f(x, y)$ have a local minimum/maximum or neither (saddle point) at $(0, 0)$?

(Recall that $D = \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$.)